

Report 2: Linear Regression

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1 Introduction

This report presents a linear regression model optimized using gradient descent to estimate parameters w_0 and w_1 .

2 Model and Loss

The model is defined as:

$$\hat{y} = w_1 x + w_0$$

he loss function used is Mean Squared Error:

$$J(w_0, w_1) = \frac{1}{2N} \sum_{i=1}^N (w_1 x_i + w_0 - y_i)^2$$

3 Implementation

The algorithm is implemented from scratch in Python. At each iteration:

- Compute gradients:

$$\frac{\partial J}{\partial w_0}, \quad \frac{\partial J}{\partial w_1}$$

- Update parameters:

$$w_0 = w_0 - r \cdot \frac{\partial J}{\partial w_0}$$
$$w_1 = w_1 - r \cdot \frac{\partial J}{\partial w_1}$$

- Print intermediate values (w_0, w_1, loss)

4 Stopping Criterion

Training stops when the change in loss is smaller than a threshold $t = 10^{-6}$.

5 Results

For lr.csv, the algorithm converges to:

$$w_0 \approx 54.40, \quad w_1 \approx 1.16$$

indicating a positive relationship between x and y . the stopping criterion is

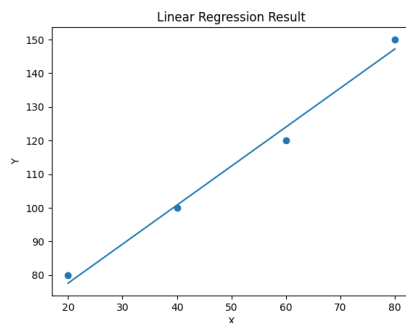


Figure 1: Regression results on two datasets

triggered when the change in loss becomes smaller than 10^{-6} , confirming convergence.

6 Effect of Learning Rate

- Small r (0.000000001): slow convergence but most stable
- Medium r (0.0001): fast and stable
- Large r (0.001): fastest may diverge (in dataset lr.csv it diverge)

7 Conclusion

Gradient descent effectively learns the parameters of linear regression. The learning rate strongly affects convergence speed and stability.